

Accumulated Transport Resistance: A Geometry-Only Surrogate for Diffusive Maintenance Burden, and a Phase-A Protocol to Test It

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Abstract

Whether an interior point of a living tissue can be maintained is a transport question, not a distance question: what matters is how much metabolic demand must be served across each available transport cross-section, integrated along the way to a supply surface. The companion paper on the mixed-boundary torsion problem establishes, with proof, that Euclidean distance to the supply set yields *no* universal lower bound on the transport burden, and that on tapered “horn” geometries a distance-only scale can overestimate the true burden by an unbounded factor. Building on that forced mathematics, we define the *accumulated transport resistance* (ATR): a demand-weighted, cross-section-weighted, path-integrated functional whose exact ground truth is the torsion field u solving $-\nabla \cdot (P \nabla u) = q$ with a Dirichlet supply portion and reflecting walls. We separate two objects the earlier work sometimes ran together—a purely geometric resistance under uniform demand and a demand-weighted concentration drop—and show they coincide only when demand is uniform. We then organize the whole problem around a single ratio, $Q_{\downarrow}(s)/(P(s)A_{\text{cross}}(s))$, the demand routed through each transport cross-section, which cleanly separates two regimes: volume-graded tapers, where the burden falls *below* the distance prediction, and demand-loaded bottlenecks, where it *exceeds* a matched-distance reference (a true-field amplification $\rho_u = 1.83$ is exhibited numerically). Because the exact field is not deployable, the scientific content is a protocol: a Phase-A “surrogate contest” on synthetic geometries with known u , in which a geometry-only ATR surrogate must pass a gatekeeper—does it track the exact torsion burden?—before any biological claim is licensed. We give the surrogate an ablation decomposition that isolates *which* ingredient carries the signal (on a matched-distance family the gain is the restriction $\times$ demand interaction, not distance, restriction, or demand alone), a validation package stronger than correlation, and a candidate eikonal construction whose error we show to be regime-dependent and largest exactly where it would be used. We close by demoting the free-energy-principle reading to an optional interpretive layer and offering, in its place, a measurable one: form responds not to raw distance but to the accumulated difficulty of maintaining relation across its own geometry. Every displayed identity and numerical claim is machine-checked by the companion harness.

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1 Introduction

A diffusing metabolite entering a tissue from a supply surface—a capillary wall, a Haversian canal, the outer boundary of a spheroid—must reach every interior cell that consumes it. The classical readout of whether it can is a *distance*: Krogh’s cylinder model [2] fixed a supply radius, and much of the downstream tradition asks only how far a cell sits from its nearest source. Distance is attractive because it is cheap: it needs only the geometry of the supply set, never a transport solve. It is also, as a predictor of maintenance burden, provably wrong in a precise and recoverable way.

The companion paper of this line [1] studies the object that actually governs the steady maintenance problem—the mixed-boundary torsion function

$$-\nabla \cdot (\mathcal{A} \nabla u) = q \text{ in } \Omega, \quad u = 0 \text{ on } \Sigma, \quad \mathcal{A} \nabla u \cdot \nu = 0 \text{ on } \Gamma, \quad (1)$$

with $\Sigma \subset \partial\Omega$ the supply (Dirichlet) portion, Γ the reflecting (no-flux) remainder, $q > 0$ a volumetric demand, and $\mathcal{A}$ a measurable, uniformly elliptic diffusivity. There u is the accumulated cost of maintaining an interior point against demand under constrained transport, and its supremum $\|u\|_\infty$ is the worst-case burden. That paper proves two things we take as given here. First, distance to Σ yields no universal lower bound on $\|u\|_\infty$: on a family of tapered horns, all three metric forms of the conjectured bound $qR^2/(2nP)$ fail, and the ratio $P\|u\|_\infty/(qR^2)$ has infimum 0. Second—and this is the point most easily misread—the failure is that distance *overestimates*: as a horn narrows, the true burden *shrinks*, while a distance-only surrogate keeps charging for the length. The corrected lesson is not “narrowing raises resistance” but something sharper, which this paper is built around:

Transport burden depends on how much maintenance demand must pass through each available transport cross-section—not on distance or narrowness in isolation.

The remainder of the paper turns that sentence into a research program. Section 2 recalls exactly what the torsion theorem forces and, importantly, what it does *not*, and it separates two transport objects that must never be multiplied by demand twice. Section 3 descends from the exact field, which is not deployable, through a PDE-derived flux diagnostic to a geometry-only surrogate, and states the discipline that governs the descent: a gatekeeper that every biological claim must pass. Section 4 introduces the organizing ratio and the two regimes it separates. Section 5 fixes two amplification numbers that the earlier draft conflated. Section 6 is the core: the Phase-A surrogate contest, its ablation decomposition, its model-comparison discipline, and a validation package that a high correlation coefficient cannot satisfy. The shorter Sections 7–10 settle the preregistration of outcomes, the supply-boundary controls, a dimensionless maintenance number, and the treatment of branch points as competing primary models rather than a sensitivity footnote. Section 11 places the lacuno-canalicular network and its controls. Section 12 demotes the free-energy reading to an optional interpretive layer and states the measurable principle that survives. Sections 13–14 record honestly what is forced, what is a modeling choice, and what remains open, and describe the machine verification. Throughout, “ P ” denotes the diffusivity scale $P_{\max}$ of (1), and we write $A_{\text{cross}}(s)$ for a transport cross-section and $V(s) = \int_0^s A_{\text{cross}}$ for the volume upstream of arclength s .

2 What the torsion theorem forces, and what it does not

2.1 The comparison field and the direction of the horn result

The horn analysis of [1] is driven by an explicit *comparison field*. On a power-law horn of profile $g(x) = \varepsilon h (x/h)^{k/2}$ with constant scalar diffusivity P , set

$$v(x, y) = a(h^2 - x^2) - b|y|^2, \quad a = \frac{q}{P(2 + k(n-1))}, \quad b = \frac{ak}{2}. \quad (2)$$

This field solves $-P\Delta v = q$ identically and—by the choice $b = ak/2$ matched to the profile exponent—has *exactly vanishing flux through the lateral wall*. It is not the torsion function: v does not vanish on the supply mouth Σ and has positive outflux on the end face. What it delivers is a rigorous one-sided comparison,

$$0 \leq u \leq v + b(\varepsilon h)^2 \quad \text{a.e.}, \quad (3)$$

hence a rigorous *upper* bound on $\|u\|_\infty$. Equation (3) is a theorem, established in the companion paper by a Stampacchia truncation against the reflecting wall; we recall it, and do not reprove it, here.

The apex value of the comparison field satisfies a graded identity that is the mathematical heart of the whole construction:

$$\frac{q}{P} \int_0^h \frac{V(s)}{A_{\text{cross}}(s)} ds = \frac{qh^2}{P(2 + k(n-1))} = v(0, 0), \quad \text{since} \quad \frac{V(s)}{A_{\text{cross}}(s)} = \frac{s}{1 + \frac{k(n-1)}{2}}. \quad (4)$$

Read (4) carefully: the tube quantity on the left *decreases* as the taper exponent k grows. Numerically it falls from 0.167 at $k = 2$ to 0.045 at $k = 10$; its derivative in k is negative throughout.

The object that *grows* without bound under narrowing is not the burden but the distance-only failure factor $qR^2/\|u\|_\infty$, which climbs from about 6 to about 22 across the same range. The horns are therefore geometries on which a distance functional and the true burden separate without bound, in the direction of distance *overcharging*. Whether the true apex value $u(0)$ equals the graded integral (4), rather than merely being bounded by it, remains open; the exact, path-free, general-domain object is instead the Green representation

$$u(x) = \int_{\Omega} G(x, y) q(y) dy, \quad (5)$$

valid wherever the mixed Green function exists, for almost every x , and pointwise under the additional regularity used in the computational phases. In the manuscript's general setting $\mathcal{A}$ may be merely measurable and uniformly elliptic, so (5) is scoped to that existence and not claimed beyond it; for the regular synthetic domains of Phase A the qualification imposes no operational limit.

2.2 Two tube objects that must not be multiplied by demand twice

A unit ambiguity haunted earlier drafts, and closing it is a prerequisite for everything downstream. There are two genuinely different one-dimensional tube integrals, and only one of them already contains demand. Define, separately, the *geometric resistance* under uniform volumetric demand,

$$\mathcal{R}_{\text{geom}} = \int \frac{V(s)}{P(s) A_{\text{cross}}(s)} ds, \quad V(s) = \int_0^s A_{\text{cross}}, \quad (6)$$

and the *predicted concentration drop* under general demand,

$$\Delta C_{\text{tube}} = \int \frac{Q_{\downarrow}(s)}{P(s) A_{\text{cross}}(s)} ds, \quad Q_{\downarrow}(s) = \int_{\text{downstream of } s} q(y) dy. \quad (7)$$

The downstream demand $Q_{\downarrow}(s)$ already carries the metabolic load, so ΔC_{tube} is the predicted drop, with the dimensions of a concentration. The geometric resistance $\mathcal{R}_{\text{geom}}$ carries only geometry. They are related, but only under uniform demand, where $Q_{\downarrow}(s) = q V(s)$ and hence

$$\Delta C_{\text{tube}} = q \mathcal{R}_{\text{geom}} \quad (\text{uniform } q \text{ only}). \quad (8)$$

Equation (8) is the whole content of the unit split: multiplying the demand-weighted ΔC_{tube} by a further factor of q would double-count the load. Keeping $\mathcal{R}_{\text{geom}}$ (volume-weighted) and ΔC_{tube} (demand-weighted) as distinct named objects removes the ambiguity at the source.

A canonical normalized burden requires a preregistered normalization Q^* : $R_G(x) = u(x)/Q^*$ is meaningful only once Q^* is fixed. For uniform demand the clean choice is $Q^* = q$, giving $R_G = u/q$ with the dimensions of $\mathcal{R}_{\text{geom}}$; total-demand and characteristic-rate normalizations are legitimate alternatives, but a specimen-specific Q^* chosen after the fact would obscure any cross-specimen comparison and is excluded unless declared in advance.

3 From exact field to deployable surrogate: three tiers and a gate-keeper

The exact torsion field u is the ground truth, but computing it requires solving the full PDE on the true geometry—precisely what a cheap predictor is meant to avoid. The program therefore

lives on three tiers. At the top is u itself. One step down is a *PDE-derived flux diagnostic*, built from the true current $J = -P\nabla u$: it records what an ideal reduction of the field would retain, but it is not deployable because it still needs u . At the bottom is a *geometry-only surrogate* R_{ATR} , computable from the segmented geometry alone. The scientific question is whether the bottom tier can be made to track the top.

The gatekeeper. We elevate one step from “validation” to a structural gate. Solving (1) on the actual geometry and testing whether a constructed R_{ATR} tracks the exact u is not a downstream nicety; it is the gate every biological claim must pass. Until that tracking is demonstrated across synthetic geometries where u is known, the honest register is

we are testing a computational surrogate inspired by the exact horn functional,

and never “we have transported the horn theorem into tissue.” The theorem is exact on its stated geometry; on an arbitrary network the surrogate is a surrogate until the gate is passed. Diffusion through irregular tissue does not descend a single path—it distributes across a field of paths—so any reduction of the field to flux tubes or watersheds is an additional modeling move, not something the theorem forces.

Flux geometry and its caveats. Because $\nabla \cdot J = q$, the current is not divergence-free: throughput varies along a tube, equal-throughput tubes need not remain equal, stream tubes encounter critical points and separatrices, a global tiling may fail or require measure-theoretic conventions, and the streamlines of J run from the interior *toward* Σ , not inward from it. The flux-tube construction is therefore a diagnostic that establishes what an ideal reduction would keep; it is not itself cheap. A deployable predictor must approximate that diagnostic *without* solving the PDE, or else be presented openly as a reduced summary of a PDE solution.

Eikonal descent is one candidate, and its error is regime-dependent. A natural cheap construction takes transport paths to be the characteristics of the eikonal equation $|\nabla T| = 1/P$, i.e. minimum-cost paths to Σ . These are not flux lines. In a controlled test the eikonal descent points straight along the supply direction (0° from the $-x$ axis, demand-blind), while the true current $-\nabla u$ curves 10.6° toward the demand mass. That single global figure understates the stakes, because the angular error between eikonal descent and the true flux *is* the modeling error incurred whenever an eikonal ATR is used, and it differs sharply between the two regimes of Section 4. Measured inside the transport corridor—the cells where the true flux is dominantly longitudinal toward Σ , about 20,000 cells per regime—the taper regime is benign (median 1.8° , ninetieth percentile 5.4°) because flux runs straight down a smoothly tapering axis, whereas the bottleneck regime has a far heavier tail (median 3.0° , ninetieth percentile 21.2° , a $3.95\times$ larger tail), concentrated exactly at the neck where flux magnitude is highest and direction matters most. The regime in which an eikonal ATR would be *used to detect* $\rho_u > 1$ is thus the regime in which it is *least reliable*. Separately, a sharp constriction produces a small flux-reversal pocket just past the neck (about 3.9% of $|J|$ at the tightest necks), where the true current briefly points back toward the interior and eikonal descent—which always points toward Σ —is about 178° wrong. This is a localized re-entrant-corner effect of classical elliptic type: it scales with corner sharpness and vanishes to zero in a straight channel, so it is a categorical rather than an angular failure, reported separately from the corridor error and not its driver. The consequences for the protocol are concrete: an eikonal ATR is admitted to the contest only with its corridor error and corner-reversal share reported per geometry, and it may not be adopted for bottleneck prediction on the

Regime	Mechanism	Effect and evidence
Volume-graded taper	downstream demand falls with cross-section fast enough	burden falls <i>below</i> the distance prediction, $\rho_u \ll 1$; power-law horns, tube quantity $0.167 \rightarrow 0.045$
Demand-loaded bottleneck	a narrow section still carries large downstream demand	burden <i>exceeds</i> a matched-distance reference, $\rho_u > 1$; a matched-Krogh bottleneck attains $\rho_u = 1.83$

Table 1: The two regimes of the transport burden. The organizing quantity is neither distance nor narrowness but the local demand-to-cross-section ratio (9).

strength of taper accuracy. The flux-tube diagnostic built from the true J , which has no such error by construction, is the reference against which every cheap surrogate is scored.

4 Two geometric regimes and the ratio that separates them

Replacing the single “narrowing raises resistance” story with the corrected mathematics produces a genuine dichotomy, and both halves are numerically real. In a *volume-graded taper*, downstream demand falls with cross-section fast enough that the burden drops below the distance prediction; the power-law horns realize it, with the tube quantity shrinking $0.167 \rightarrow 0.045$ as computed in (4). In a *demand-loaded bottleneck*, a narrow section still carries large downstream demand, and the burden can exceed a matched-distance reference. Table 1 records the two.

The quantity that organizes both is the *local ratio*

$$\frac{Q_{\downarrow}(s)}{P(s) A_{\text{cross}}(s)}, \quad (9)$$

the maintenance demand routed through each available transport cross-section. A narrow section is cheap when little demand remains beyond it; a modest bottleneck is severe when it must serve a large downstream region. This is a stronger and more falsifiable statement than the original distance conjecture, and it is what Phase A is built to test. The existence of the $\rho_u > 1$ regime is not hypothetical: a bottleneck matched to its Krogh reference on maximum distance, total demand, and volume attains a true-field amplification $\rho_u = 1.83$. The *magnitude* of achievable amplification, and whether a closed-form bottleneck family exists, remain the sharpest open targets of the program.

5 Two amplification ratios, kept distinct

An earlier draft wrote a single symbol ρ for two different things, and the resulting confusion is worth preventing permanently. The *true-field amplification* compares the exact burden to that of the matched Krogh reference geometry,

$$\rho_u = \frac{\|u\|_{\infty}}{u_{\text{Krogh}}}, \quad (10)$$

so $\rho_u > 1$ means the geometry is more burdensome than its Krogh reference, $\rho_u < 1$ less, $\rho_u = 1$ neither. Horns have $\rho_u \ll 1$; the demand-loaded bottleneck has $\rho_u > 1$. The *surrogate amplification* is the same comparison computed from the deployable surrogate,

$$\hat{\rho}_{\text{ATR}} = \frac{R_{\text{ATR}}^{\text{sur}}}{R_{\text{ATR,Krogh}}^{\text{sur}}}. \quad (11)$$

Phase A asks precisely whether $\hat{\rho}_{\text{ATR}}$ predicts ρ_u . The two must be kept apart exactly because the horn family lives at $\rho_u \ll 1$ while the bottleneck family lives at $\rho_u > 1$: a single word “amplification” spanning both would be actively misleading.

6 The Phase-A surrogate contest

The scientific object of this paper is not a biological result but a protocol whose first phase settles a mathematical question with no biology in it at all: *which* geometry-only summary, if any, tracks the exact torsion burden.

The four phases. Phase A runs the contest on synthetic domains matched on maximum Euclidean distance, volume, total demand, diffusivity, and supply-boundary geometry (Section 8), with cross-sectional restriction varied systematically across *both* regimes. For each geometry we compute and rank against the exact $\|u\|_\infty$ a ladder of candidates: Euclidean maximum distance; geodesic distance; a skeleton or medial-axis ATR; an eikonal ATR; the flux-tube diagnostic from the true J ; and the exact field itself. Phase B applies the survivors to real segmented geometries and confirms calibration against the full PDE on the same geometry—the gatekeeper again—with no outcome variable yet. Phase C introduces a matched biological readout and tests incremental predictive power beyond distance and the standard covariates, with the outcome-to-functional mapping preregistered (Section 7). Phase D asks the developmental question—do organisms alter architecture in the predicted direction—and is reached only after the physical predictor has passed; cross-sectional tissue data can show association only, so the threshold-activation claim ultimately wants developmental or perturbational evidence.

An ablation ladder, so no surrogate is a black box. Reporting only a surrogate’s final score against u would leave unknown *which* ingredient produced it. Every surrogate is therefore reported along a fixed ladder that adds one information source at a time. The first rung uses Euclidean distance only; the second replaces it with geodesic distance; the third adds the local cross-section, i.e. restriction; the fourth instead adds downstream demand; the fifth is the full surrogate, in which restriction and demand are combined multiplicatively and integrated along the path; and the sixth rung is the exact PDE burden u , the ceiling. For each rung we report the same package used everywhere in the protocol—grouped cross-validated R^2 , median relative error, Spearman correlation, calibration slope, and matched-pair ordering—so that the question “which rung produces the gain” has a quantitative answer that is itself a result, independent of whether the final surrogate is ever adopted.

The ladder is decidable, and the decomposition is stark. A ladder is only meaningful if its rungs actually separate; two collinear adjacent rungs would make the decomposition illusory. On a 16-geometry family spanning both regimes (8 tapers and 8 demand-loaded bottlenecks, all matched on maximum distance), the largest adjacent-rung collinearity is $|\text{corr}| = 0.369$ —far below

the ≈ 0.98 that would signal redundancy, so the rungs carry independent information. On that matched family the decomposition is decisive. Because maximum distance is held fixed, Euclidean distance is blind (corr = 0.00 with u); cross-sectional restriction alone is partial and *inversely* aligned (-0.85); downstream demand alone is only weakly aligned ($+0.23$); and only their path-integrated product tracks the true burden ($+0.92$). The predictive gain is neither geometry nor demand in isolation but the restriction $\times$ demand interaction—a concrete, pre-biological instance of the thesis that burden tracks $Q_{\downarrow}(s)/(P(s)A_{\text{cross}}(s))$. Whether this decomposition survives on real segmented geometries is itself a Phase-B question.

Where a rung’s path or direction is built from an eikonal field, its error is not uniform across the family, for the reasons of Section 3: small in the taper regime but with a roughly fourfold heavier tail in the bottleneck, plus the re-entrant-corner reversal that eikonal cannot represent. Each eikonal-dependent rung is therefore reported separately by regime, so that a rung which looks adequate on the taper-dominated average is not credited in the bottleneck regime it exists to detect.

Four models and the discipline of comparing them. The biological test of Phase C is organized around four nested-or-not models,

$$\begin{aligned} M_0 &: \text{outcome} \sim \text{distance}, \\ M_1 &: \text{outcome} \sim \text{distance} + \text{demand} + \text{diffusivity} + \text{volume} + \text{covariates}, \\ M_2 &: M_1 + R_{\text{ATR}}^{\text{SURR}}, \\ M_3 &: M_1 + R_G \quad (\text{the exact-}u \text{ summary}). \end{aligned}$$

The comparisons must respect nesting. M_1 versus M_2 and M_1 versus M_3 are nested and are compared by likelihood-ratio tests together with grouped-CV ΔR^2 and AIC/BIC. But M_2 versus M_3 compares a surrogate against the exact burden and is *generally non-nested*: it must be compared by grouped cross-validation, information criteria, or Vuong-type tests, never by a likelihood ratio. Crucially, “ $M_2 \approx M_3$ ” is not a discretionary judgment; we declare *surrogate parity* only when M_2 ’s grouped out-of-sample performance falls within a preregistered tolerance of M_3 —for instance $\Delta R_{\text{CV}}^2 \leq 0.02$ with comparable calibration slope and matched-pair ordering. The interpretation then separates two failure modes that a single test would confound: if $M_3 > M_1$ but $M_2 \not\approx M_1$, the physical burden matters and the surrogate failed; if neither beats M_1 , modeled burden adds nothing in that system; and if $M_2 \approx M_3 > M_1$, burden matters and the surrogate captures it—the target.

Why a high correlation is not enough. A surrogate can correlate almost perfectly with the truth and still invert the comparisons that matter. In a direct construction, a surrogate with Pearson correlation 0.995 against u nonetheless reverses the ordering of 20 matched-distance pairs. The minimum acceptance package is therefore not a correlation but the full set: cross-validated R^2 , median relative error, Spearman correlation, calibration slope and intercept, and above all *pairwise-ordering preservation at matched distance*, together with out-of-sample performance and behavior under severe bottlenecks. The surrogate must beat maximum distance on geometries constructed so that the two disagree—not on a sample where distance already happens to work.

Biological outcome	Matched functional of u
local osteocyte viability	$u(x)$, or a local-neighborhood summary
total necrotic fraction	the measure of $\{x : u(x) > \Delta C_{\text{avail}}\}$
mean oxygenation	a spatial or demand-weighted mean of u
architectural density	a percentile, or integrated exposure $\int \mathbf{1}[u > \theta]$

Table 2: Preregistered mapping from the PDE summary to the biological outcome. Choosing the summary after seeing the data would invalidate the incremental- predictivity test of Phase C.

7 Matching the summary to the outcome

The scalar $\|u\|_\infty$ is the right summary for a specimen-level failure threshold and the wrong one for a local or averaged outcome; the mapping from PDE summary to biological readout must be fixed before Phase C, not chosen to fit. Table 2 records the preregistered correspondence.

8 Supply-boundary geometry as a matched variable

A clean bottleneck experiment must hold fixed not only the maximum distance but the effective supply interface, so that interior cross-sectional restriction is the sole varied factor. Phase A therefore controls or explicitly varies the supply set Σ : its total Dirichlet surface area, the number of disconnected supply patches, their placement, the boundary concentration, and—if a Robin contact condition later replaces ideal Dirichlet—the contact conductance. The matched-Krogh bottleneck that attains $\rho_u = 1.83$ was matched to its reference on maximum distance, total demand, and volume for exactly this reason; without the supply interface held fixed, an apparent amplification could be an artifact of a changed boundary rather than of interior geometry.

9 A dimensionless maintenance number

With the unit split of Section 2 in hand, the feasibility of maintenance admits a clean dimensionless form. Writing C_Σ for the supply concentration and C_{crit} for the viability threshold, set

$$\Pi_{\text{maint}} = \frac{\Delta C_{\text{tube}}}{C_\Sigma - C_{\text{crit}}}, \quad \text{and, for uniform } q, \quad \Delta C_{\text{tube}} = q \mathcal{R}_{\text{geom}}. \quad (12)$$

Then $\Pi_{\text{maint}} < 1$ is viable, $\Pi_{\text{maint}} \approx 1$ is threshold, and $\Pi_{\text{maint}} > 1$ predicts depletion beyond viability. The correction that makes this honest is dividing by the *margin above threshold* $C_\Sigma - C_{\text{crit}}$, not by C_Σ alone. One should report Π_{maint} alongside the true-field amplification ρ_u of (10), and never alongside the comparison-field quantity, which shrinks with taper and would understate the load.

10 Branch points: competing primary models, not a footnote

At a branch point the upstream demand must be split among daughter paths in order to evaluate $Q_\downarrow(s)/A_{\text{cross}}(s)$, and there are two natural rules. An *area-proportional* split sends demand in proportion to the downstream cross-section carried by each daughter, in the spirit of Murray’s law

and a Kirchhoff balance [3]; a *demand-proportional* split sends it in proportion to the downstream integrated demand of each subtree. In the real PDE the branch flux is set jointly by conductance, geometry, sources, and gradients, and neither shortcut is automatically the biological rule. We therefore run both as *competing primary models*, not as a sensitivity appendix, and report their full model comparisons side by side. If the ATR surrogate wins under both rules, the result is robust and partition-independent; if it wins under only one, that is informative about *what* the functional detects—whether demand-weighting or area-weighting carries the signal—rather than a nuisance to be hidden.

11 The lacuno-canalicular test case and its controls

The lacuno-canalicular network (LCN) of bone is the strongest biological discriminator available, because its measured, restricted cross-section departs sharply from Krogh distance; it is exactly where distance and ATR most disagree. For that same reason it must *not* be the first implementation. An LCN result carries meaning only after the surrogate has passed the gatekeeper on controlled synthetic horns and branching networks where the true u is known.

Even then, a specimen-level LCN test discharges only under conditions that the protocol states in advance. The supply set may exceed the Haversian canal, drawing on Volkmann canals, neighboring osteons, bone and marrow surfaces, and pressure gradients. Transport may include load-induced advection, requiring $-\nabla \cdot (P\nabla c) + \mathbf{v} \cdot \nabla c = q$ rather than pure torsion. Canalicular density cannot be both predictor and outcome—it constructs the cross-section A_{cross} , so the dependent variable must be independent of it (osteocyte viability, lacunar occupancy, recovery time, solute penetration, or metabolic markers). And radial averages erase the very bottlenecks, connectivity, and dead-ends that the functional is meant to detect, so a confirmatory test needs specimen-level three-dimensional geometry paired with matched specimen-level readouts, not a mean profile paired with unrelated viability data. Passing all four, cross-sectional data still show association only; the threshold-activation claim wants the developmental or perturbational evidence of Phase D.

Two controls frame the discriminator. *Spheroids* are a principled near-Krogh negative control, with $\rho_u \approx 1$ expected; a null result there confirms the domain of relevance rather than refuting the concept. *Muscle* is a stress test rather than a confirmation: myoglobin-facilitated transport and the mitochondrial reticulum can negate classical diffusion distances, so both ATR and Krogh distance may fail there, and this should be flagged prominently because muscle-literate readers will probe it.

12 Interpretation: an accessibility potential, and the demotion of the free-energy reading

The empirical content of this program stands with the free-energy principle (FEP) discarded. FEP is, at most, an optional and terminal interface language for maintenance under uncertainty; it does not generate the transport functional, and the explanatory arrow stays physical: geometry sets the PDE burden $R_G = u/Q^*$, which sets maintenance feasibility. We state this demotion plainly so that no predictive weight rests on FEP.

What survives, and is measurable, is a reading of u as an *accessibility- cost potential*: the scalar cost of maintaining an interior point under a supply boundary, distributed demand, and constrained transport geometry, with R_{ATR} its pathwise accumulation. The relation is clean—

R_{ATR} rising is accessibility falling. When passive accessibility becomes too costly, a system can change geometry, widen cross-section, alter permeability, add sources, or grow active- maintenance architecture. This is a material realization of a single principle, offered as interpretation of the same physics rather than as an independent source of predictions:

Form does not respond to raw distance. It responds to the accumulated difficulty of maintaining relation across its own geometry.

Unlike the FEP bridge, this reading asserts no upstream awareness or optimization; it supplies a middle layer from relational geometry to biological architecture that is, in principle, measurable.

13 Status of the claims

Because the protocol mixes theorems, modeling choices, and open questions, we record which is which without decoration.

Forced by the current mathematics. The Green representation (5) holds where the mixed Green function exists (a.e. x , pointwise under Phase-A regularity). Distance to the supply set yields no universal lower bound. The explicit horn comparison field (2) solves the PDE with exactly vanishing lateral-wall flux, and its apex obeys the graded identity (4); the field gives a rigorous upper bound (3) on the true burden. A distance-only burden can overestimate the true torsion by an arbitrarily large factor on the horn family (failure factor $6 \rightarrow 22$). For uniform demand, $\Delta C_{\text{tube}} = q \mathcal{R}_{\text{geom}}$ (8). A demand-loaded bottleneck can be more burdensome than its matched Krogh reference, $\rho_u = 1.83$, establishing the $\rho_u > 1$ regime. On a matched-distance family spanning both regimes the ablation rungs separate and the predictive gain is the restriction $\times$ demand interaction, not distance, restriction, or demand alone (correlations with u of 0.00, -0.85 , $+0.23$, $+0.92$). The eikonal-surrogate error is regime- dependent—small in the taper regime, with a roughly fourfold heavier tail in the bottleneck, plus a re-entrant-corner flux-reversal pocket the eikonal cannot represent—so it is least reliable in the very regime it would be used to detect.

Declared constructions. The distinct objects $\mathcal{R}_{\text{geom}}$ and ΔC_{tube} ; the normalized burden $R_G = u/Q^*$; the flux-tube diagnostics; the eikonal, skeleton, and geometry-only surrogates; the area- and demand-proportional branch partitions; the maintenance number Π_{maint} ; the amplifications ρ_u and $\hat{\rho}_{\text{ATR}}$; and the accessibility-potential reading are all modeling choices, sound but not forced.

Open. Whether the true horn apex $u(0)$ equals the graded tube integral rather than merely being bounded by it; the magnitude of achievable $\rho_u > 1$ and any closed-form bottleneck family; whether a uniquely preferred general-domain surrogate exists; whether ATR beats distance plus conventional covariates on real biology; whether a universal persistence envelope exists; and any FEP derivation.

14 Computational verification

Every displayed algebraic identity and rational inequality above, and every numerical claim about the two regimes, is machine-checked by the companion harness `surrogate_contest.py` (exit 0), which builds each synthetic geometry, solves (1) to obtain the exact u , and scores the candidates. The eight check families are: (C1) eikonal descent differs from the true flux, 10.6° , with eikonal 0° from the supply direction; (C2) a matched-distance, matched-demand bottleneck raises the true $\|u\|_\infty$ by $1.66\times$, tracked by the exact tube reduction at $1.63\times$ —distinct ratios—and missed by geodesic distance (1.6%); (C3) a Pearson-0.995 surrogate inverts 20 matched pairs; (C4) the

horn comparison quantity shrinks with taper ($0.167 \rightarrow 0.045$) while the distance-only failure factor grows ($6 \rightarrow 22$); (C5) the uniform-demand reduction $\Delta C_{\text{tube}} = q \mathcal{R}_{\text{geom}}$ holds with no double-count; (C6) a demand-loaded bottleneck attains $\rho_u = 1.83 > 1$ against its matched Krogh reference—the open regime, numerically realized; (C7) the ablation ladder is decidable (largest adjacent-rung $|\text{corr}| = 0.369$) and mechanism-isolating (the gain is the restriction $\times$ demand interaction, with correlations $0.00 / -0.85 / +0.23 / +0.92$); and (C8) the eikonal-surrogate error is regime-dependent (taper corridor ninetieth percentile 5.4° versus bottleneck 21.2°) with a small re-entrant-corner reversal pocket at sharp necks. The corner-reversal effect was probed for standalone significance by a second companion script, `reversal_generality.py`: it is generic and resolution-independent but a local elliptic-corner phenomenon that scales with neck sharpness and vanishes in a straight channel—classical, not novel—and is folded in as a caveat rather than a result.

15 Outlook

The logical spine of the program is short and depends on no equality the horn paper did not establish:

exact torsion field $u = \int G q \rightarrow$ comparison-field horn identity $v(0,0) = \frac{q}{P} \int V/A_{\text{cross}}$ (an upper bound on u) $\rightarrow$ geometry-only surrogate contest across both regimes (Phase A) $\rightarrow$ real-geometry PDE calibration, the gatekeeper (Phase B) $\rightarrow$ biological-outcome test with the four models and both branch partitions (Phase C) $\rightarrow$ a possible persistence principle, developmentally probed (Phase D).

The highest-value next step is Phase A: to determine which general-domain surrogate, if any, tracks the exact torsion burden across both the taper and the demand-loaded-bottleneck regimes under adversarial perturbation—with the bottleneck family, now numerically real at $\rho_u = 1.83$, as the sharpest probe.

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